



The effect of group walking program on social physique anxiety and the risk of eating disorders in aged women: A Randomized Clinical Trial study

Sharare Khalili^{a,b}, Fatemeh Shirinkam^a, Reza Ghadimi^c, Hengameh Karimi^{a,*}

^a Nursing Care Research Center, Ramsar Nursing and Midwifery School, Babol University of Medical Sciences, Babol, Islamic Republic of Iran

^b Mazandaran University of Medical Sciences, Sari Imam Khomeini Hospital, Sari, Mazandaran, Islamic Republic of Iran

^c Social Determinants of Health Research Center, Health Research Institute, Babol University of Medical Sciences, Babol, Islamic Republic of Iran

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ABSTRACT

Objective: The aim of the Randomized Clinical Trial (RCT) study was to determine the effect of Group Walking Program (GWP) on social physique anxiety (SPA) and the risk of Eating Disorders (EDs) in aged women.

Methods: 70 young-old women with medical records in Sari-Iran health centers were selected by a multi-stage sampling. They obtained score of 15 or more from SPA Scale (SPAS), a score of 20 or more from Eating Attitude Test (EAT-26). The group walking lasted eight weeks. 70 young-old women with medical records in Sari-Iran health centers were selected by a multi-stage sampling. They obtained score of 15 or more from SPA Scale (SPAS), a score of 20 or more from Eating Attitude Test (EAT-26). The group walking lasted eight weeks. To compare differences in EDs and SPA at post-test, analysis of Covariance was used.

Results: The results showed significant differences in the mean score of SPA ($P = 0.01$), EDs ($P < 0.001$), and EDs subscales between the groups.

Conclusion: The GWP can be used as an easy and affordable solution to reduce SPA and EDs in the aged women.

1. Introduction

Some aged people are concerned with their physical appearance, weight and its control as well as how the community views them (Homayounnia Firoozjah et al., 2015). Most people worry about how their body would look like from outsiders' perspective. In social encounters, it's the social physique anxiety (SPA). SPA is considered as one of the people's concerns. This anxiety is experienced when people are worried about others' evaluations on their body. The SPA is one of the aspects of social anxiety related to dissatisfaction with the physical condition, and people understand it when they think that other people evaluate their body negatively (KHajavi & GHanbari, 2013). The SPA plays an important role in health-related behavior and knowledge (Jajat & Hardwis, 2020). Ransdell et al. (2008) reported that the mean SPA in the United States was 27.3% and 26.9% for women aged 60–69 and 70–79, respectively (Ransdell et al., 2008). According to Rothberger, exercise and proper physique are two important indicators in preventing SPA (Rothberger, 2014). Akyüz has suggested that the eight-week sport program has a positive effect on SPA (Akyüz, 2017). On the other hand, the elderly who do sport activities may also have some degree of eating

disorders (EDs) which can appear as obsessive about their physical appearance (Homayounnia Firoozjah et al., 2015). The EDs include abnormal attitudes toward current and ideal weight, mental image of the body and eating behaviors, mental preoccupation with food, their metabolism in the body and methods of eliminating food from the body, and eventually EDs may lead to severe clinical conditions such as anorexia nervosa and bulimia nervosa (Hrabosky & Grilo, 2007; Pourghassem Gargari et al., 2010). The EDs have potential and serious physical and psychological consequences (Koushiou et al., 2020). The results of some studies show that the SPA (Murphy, 2012) and EDs are more common in women than men (Mohammadzadeh et al., 2013; Stewardson et al., 2020). The prevalence of EDs based on Diagnostic and Statistical Manual-5 (DSM-5) has been reported 3.5% in aged women and 1–2% in aged men (Smink et al., 2012). The literature on EDs and SPA in elderly women is very limited. In Iran, although the researchers have not succeeded in finding the prevalence of the variables in aged people, the results of a pilot study on 20 elderly people who met the inclusion criteria demonstrated that they had a 60% risk of EDs and SPA. Women with EDs are more likely to seek weight control than others and are anxious about their future, and this anxiety encourages them to do

* Corresponding author at: Ramsar Nursing Care Research Center, Ramsar Nursing and Midwifery School, Babol University of Medical Sciences, Babol 46917-14141, Islamic Republic of Iran.

E-mail address: hengamehkarimister@gmail.com (H. Karimi).

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exercise and physical activity as the easiest way to balance the weight (Burls, 2007). Regular physical activity plays an essential role in providing physical and mental health in all periods of life, especially aging period (Razaghi et al., 2020). In the meantime, walking as one of the sports is very safe and easy for most aged people, does not require any special equipment except proper shoes and can easily be done indoors or outdoors, alone or in groups (Brunner, 2010). Physical activity as a group rather than as individuals has a greater effect on the mental health of the elderly (Mortazavi et al., 2012). Outdoor group walking is physically and mentally beneficial for participants (Hanson et al., 2016) and causes to stimulate and create idea (Oppezzo & Schwartz, 2014), increase self-confidence (Peacock & Pretty, 2007), decrease stress, improve mood and quality of life (Pretty et al., 2007), improve sleep quality (Farshadbakht et al., 2015) and treat sleep problems (Rahmaninia et al., 2010).

Because in contemporary Iran, having a beautiful physique is especially important among women and academic studies addressing discourses the effect of group walking on the risk of EDs and SPA in the elderly in Iran are scarce, the aim of the present study was to determine the effect of group walking program on the risk of EDs and SPA in aged women. The first objective was EDs and SPA changes on the group walking program and the secondary objective was to determine the effect of the program on EDs subscales and BMI. Our hypothesis was that group walking program would positively effect on the risk of EDs and SPA, also on EDs subscales and BMI in aged women.

2. Materials and methods

The Randomized Clinical Trial was down without blinding. The research setting was health centers of Sari due to the easy access to the participants. The research was carried out in summer 2018. Participants were 70 young-old women (60–74 years old) with medical records in Sari health centers.

2.1. The inclusion criteria

The inclusion criteria were: being 60–74 years old, having no known chronic illnesses according to health records, having an interest and satisfaction to participate in research (obtaining informed consent), having no history of acute events and incidents leading to mental and physical problems over the past month (Heidari et al., 2013), taking no special drugs (Antihypertensive drug, blood sugar and heart medications, etc.), obtaining a score of eight or more from Abbreviated Mental Test (AMT), obtaining an independent score from questionnaire of Activities of Daily living (ADLs), obtaining a score of 15 or more from social physique anxiety scale (SPAS), obtaining a score of 20 or more from eating attitude test-26 (EAT-26) and having no record of regular exercise program during the last six months (McAuley et al., 2002).

2.2. The exclusion criteria

The exclusion criteria were: reluctance to continue participating in the study, absence in more than three consecutive sessions in walking sessions, incidence of unexpected events as well as severe mental and physical crises during the study, hospitalization during the study and intolerance of walking for 30 min.

2.3. Sample size

Based on a previous study (Pahlavan Yali & Hojjati Zidashti, 2016) using Eq. (1) and regarding test power at the level of 10% = 1.28 ($Z_{1-\beta}=1.28$), $S_1^2 = 9.8$, $S_2^2 = 10.5$, $\mu_1 = 27.9$ and $\mu_2 = 21.9$, the sample size was 60 (30 per group). Considering a drop-out rate (16%), the sample size required 70 participants (35 per group).

$$n = \frac{\left(Z_{1-\frac{\alpha}{2}} + Z_{1-\beta} \right)^2 (S_1^2 + S_2^2)}{(\mu_1 - \mu_2)^2} \quad (1)$$

$$n = \frac{(1.96 + 1.28)^2 (9.8 + 10.5)^2}{(27.9 - 21.9)^2} = 60$$

The sampling was multi-stage. The first stage was cluster sampling. So that each of the 12 health centers located in Sari City was considered as one cluster, then among the clusters, six clusters were randomly selected, and then from each cluster, the aged women who met inclusion criteria were selected with systematic randomized sampling, proportional to the sample size and based on the list of names of the elderly to the Electronic Health Record available in the Integrated Health System (IHS) which in Persian is called SIB. They were divided into two experimental and control groups with randomized allocation and matched in terms of age and BMI. Primary outcome of this study was SPA and EDs changes on three sessions a week of eight weeks group walking program. Secondary outcome was to determine the effect of the program on EDs subscales and BMI. Data were collected using demographic and ADLs questionnaires, AMT, SPAS, EAT-26 and SIB records. Demographic questionnaire includes age, marital status, educational level, place of residence, home-ownership status, health insurance coverage and supplementary insurance status, cohabitation status, financial status, family's income adequacy and occupational status.

2.3.1. ADLs questionnaire

This tool consists of eight items including personal cleaning, urinary and fecal control, eating, dressing, moving, walking, bathing and toileting. Each item has three options of “dependent” (score = 0), “needs help” (score = 1) and “independent” (score = 2). Total score of this indicator is from zero to 16 and each subject is placed in one of the three categories of “dependent” (a score from 0 to 7), “needs help” (a score from 8 to 11) and “independent” (a score from 12 to 16). In Iran, the content validity index for this questionnaire is 0.82 and Cronbach's alpha is reported 0.80. This questionnaire has been confirmed outside of Iran too (Taheri Tanjani & Azadbakht, 2016).

2.3.2. AMT

This test consists of 10 items. Obtaining a score of eight or more is required to participate in the study. The validity and reliability of this questionnaire in were verified with Cronbach's alpha coefficient of 0.76 for internal reliability and 0.89 for external reliability for a good tester (Bakhtiyari et al., 2014).

2.3.3. SPAS

The SPAS of Motl and Conroy (2000) consists of seven items provided based on a primary 12-item scale of Hart et al. (1989) and measures a person's anxiety level about the shape of his/her body in social situations (Hart et al., 1989; Motl & Conroy, 2000). The subjects determine their response to each item on a Likert scale from 1 (not at all) to 5 (very extremely) appropriate to their situation. Item 5 has a reverse score in this questionnaire. The score range is from 7 to 35 (Poursharifi et al., 2014). Higher scores indicate more SPA. In Iran, the reliability of the test was 0.85 for women and 0.81 for men who are students (Yousefi et al., 2009) as well as was 0.94 in the study of Scott et al. (2004). In this study, the face validity of this questionnaire was confirmed by 10 faculty members of Ramsar School of Nursing and Midwifery and its reliability was confirmed by test-retest on 15 samples with Cronbach's alpha of 89%. These 15 people were excluded from the ongoing study.

2.3.4. EAT-26

The main form of this questionnaire consists of 40 items. Through factor analysis, Garner et al. (1982) clarified that 14 items of the

questionnaire were not necessary for any of the cases and the questionnaire of 26 questions with three subscales was obtained. The first subscale is “dieting”, which is related to avoiding fattening foods and mental preoccupation with being thinner. The second subscale of “bulimia and food preoccupation” comprises cases that show thoughts associated with food and, in the same proportion, binge eating. The third subscale of “oral control” relates to the self-control of eating and perceived pressure from others to gain weight. Each of the items in this questionnaire is graded on a Likert scale. For each phrase, the answers of “always”, “very often” and “often” get a score of 3, 2 and 1, and the remaining three options including “sometimes”, “rarely” and “never” receive a score of zero, respectively. The item 25 has reverse score. The scores can range from zero to 78. If the answer is graded as “never”, “rarely” or “sometimes”, it is not considered important, whereas if it is graded as “always”, “very often” and “often”, it can be a sign of a clinical problem. Obtaining a score of 20 or more from this questionnaire is defined as a risk of eating disorders, and a higher score indicates likely a higher risk of EDs (Garner et al., 1982).

In a study, Cronbach's alpha coefficient was 0.75% on Brazil (Nunes et al., 2005), and in other studies were 0.91 (Nobakht & Dezhkam, 2000) and 0.76 (Pourghasem Gargari et al., 2009) in Iran. The face validity of this questionnaire was confirmed by 10 faculty members of Ramsar Nursing and midwifery school. The reliability of this tool in this study was confirmed by the test-retest method on 15 subjects with Cronbach's alpha of 81%. These 15 elderly women were excluded from the present study.

2.3.5. SIB

SIB records were used to determine BMI. The health caregivers at each health center, based on the patient's follow-up status displayed on the SIB, announced the next visit time to the patient, which was every three months, on average.

2.4. Procedure

The aged women were present in health centers after the phone call on specific days. In the same days, the researcher referred to these centers and took samples based on the inclusion criteria. Necessary explanations about the purpose of the research, method of completing the questionnaires, freedom of the participants to attend in the study or withdraw from the current study, conditions of the walk, time, number and duration of the walking sessions as well as confidentiality of information were provided for the research centers, and the informed consent was obtained from the elderly women. The questionnaires were then completed by the researcher in a face-to-face interview. After reviewing the questionnaires, the aged women were entered into the present study based on inclusion criteria. For each of the elderly, a number from one to 70 was randomly selected. Even numbers to experimental group and odd numbers to control group were allocated. Participants in experimental group were invited by a reminder telephone call and short message service (SMS) to attend the walking sessions.

2.5. Intervention

The intervention was carried out in Sari Mellat Park. This park has a suitable and beautiful space for walking. Hekmat Hospital is the nearest hospital to Mellat Park, and it was expected that if there was a physical problem for the participants during the walk, they could be transferred to this center. The group walking program lasted three sessions a week for eight weeks, with the participation and supervision of one of the researchers who was constant during the sampling and intervention. The duration of each walking session was 30 min (Karimi Torghabeh et al., 2011; Mortazavi et al., 2012). Due to the warm weather, the program was held in the evening (19 to 19:30 p.m.). The aged women were recommended not to participate in any other sports or walking program during the intervention, and the cost of transportation was paid to the

aged women who lived far from the park from the first session. The aged people in appropriate shoes and clothing would attend the sessions on time, and if someone got tired during walking, they would rest for a while and then join the group. In order to encourage aged persons to participate regularly in the intervention, it was announced that in the last walking session, a prize would be awarded by drawing lots to one of the women who regularly attended the intervention sessions. At this stage, no intervention was performed for the control group. The experimental group was 35 aged people, and four of them withdrew from the study due to the absence of more than three consecutive sessions in walking sessions (two women due to travel, one owing to illness and one because of working on the farm). Totally, 31 elderly women participated in all walking sessions.

In the last session, after the intervention, both SPAS and Eat-26 were completed with face-to-face interview again as post-test. Post-test time was determined with regard to similar studies (Akyüz, 2017; Mollazadeh Esfanjani et al., 2012). The number of the elderly in the control group in post-test was 31 persons, and four women withdrew from continuing cooperation due to travelling. In addition, the results of the intervention in the experimental group were informed to control group via SMS and phone call, and the necessary explanations were provided for them about how to implement walking program. These results were also provided to the experimental group to strengthen the motivation for continuing the walking program.

2.6. Moral considerations

During this research, the researchers required to observe the following cases: obtaining a license from the Ethics Committee of Babol University of Medical Sciences (MUBABOL. HRI. REC.1396. 123), obtain a license from the RCT of Iran (IRCT20180505039538N1), getting the introduction letter from the school, Mazandaran Health Network, Mazandaran Research and Technology Center and the city health center. Introducing researchers to the participants and mentioning the research goals to them, getting informed consent from the elderly, freedom of participants to participate in the study or withdraw from the current study, ensuring confidentiality of information as well as providing research results to the Research Deputy of Babol University of Medical Sciences, research officials as well as control and experimental groups.

2.7. Data analysis

In the present study, descriptive statistics were used to classify the test scores as well as calculate the frequency, mean and standard deviation, analytic statistics to examine the success of randomization procedures (chi-square and Fisher's exact test) were used. To compare differences in EDs, SPA and BMI, Analysis of Covariance (ANCOVA) and to compare differences in EDs subscales in the post-test, Multivariate Analysis of Covariance (MANCOVA) were used. We used the Walking program groups (Experimental/Control) as a fixed factor, EDs and SPA in the pretest as a covariate. Levene's test, Kolmogorov-Smirnov test were carried out and the other assumptions of ANCOVA were met. Analysis of Covariance was used (ANCOVA). Statistical analyses were used with SPSS v26. The significance level was considered 0.05.

3. Results

Participant flow is shown in Fig. 1. Of the 100 individuals who were assessed for eligibility, 70 participants met inclusion criteria and were allocated into two groups.

3.1. Subject characteristics

The mean age of the experimental and control groups was 64.2 ± 3.1 and 64.2 ± 2.6 years, respectively and the mean BMI in both groups was 27.5. Characteristics of the groups are shown in Table 1. In both groups,

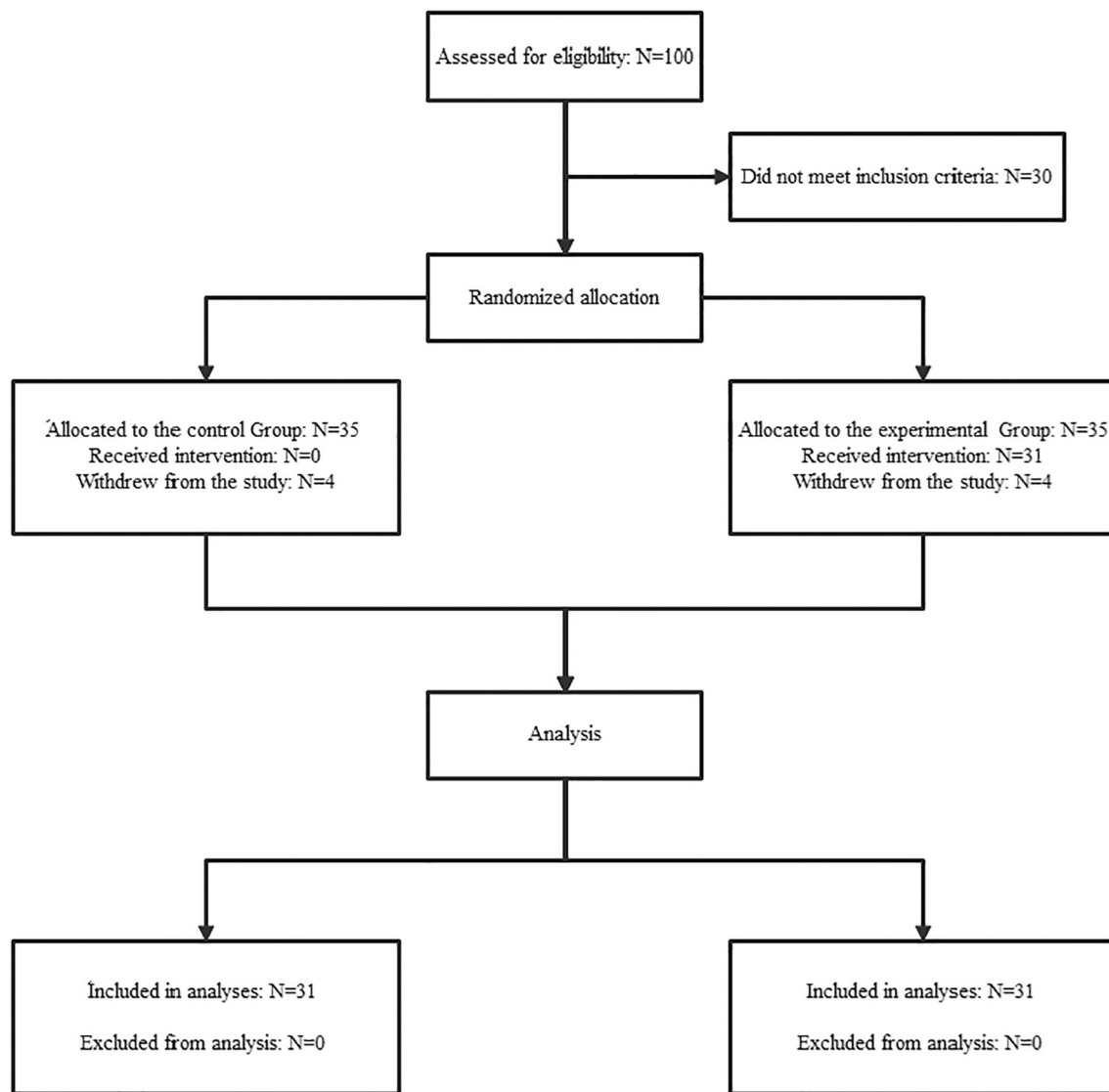


Fig. 1. Flow chart of the study and participants.

one of the women remarried and all participants were living in the city, having a private home and covered by health supplementary insurance. There were no significant differences between the two groups on age, BMI and any of the variables described in this demographics section.

3.2. SPA

The mean score of SPA before and after the intervention in the experimental group was 17.7 ± 3.2 and 14.6 ± 2.9 and in the control group was 16.5 ± 3.1 and 16.2 ± 2.5 respectively. As shown in Table 2, there was a significant difference in mean scores of SPA ($F = 7.6$, $p = 0.008$) between the groups. The result indicated the intervention had effected on SPA and the walking group program reduced SPA in the experimental group.

3.3. EDs

The mean score of EDs before and after the intervention in the experimental group was 29.2 ± 4.5 and 21.1 ± 3.8 and in the control group was 29.8 ± 4.7 and 28.7 ± 5.5 respectively. As shown in Table 2 there was a significant difference in mean scores of EDs ($F = 45.84$, $P < 0.001$) between the groups.

3.4. EDs subscales

The mean scores of dieting, before and after the intervention in the experimental group were 15.1 ± 2.8 and 10.9 ± 2.4 and in the control group were 15.4 ± 3.03 and 14.7 ± 3.9 respectively. As shown in Table 3, there was a significant difference in mean scores of dieting ($F = 29.89$, P) between the groups. The result indicated the intervention had effected on dieting and the walking group program reduced dieting in the experimental group.

The mean scores of bulimia and food preoccupation before and after the intervention in the experimental group were 6.5 ± 1.7 and 4.6 ± 1.8 and in the control group were 6.1 ± 1.6 and 5.9 ± 1.4 respectively. There was a significant difference in mean scores of the subscale ($F = 7.68$, $p = 0.007$) between the groups (Table 3).

In the oral control subscale, the mean scores before and after the intervention in the experimental group were 7.5 ± 2.3 and 5.7 ± 1.8 and in the control group were 8.2 ± 1.4 and 8.06 ± 1.4 respectively. As shown in Table 4–4, there was a significant difference in mean scores of oral control ($F = 36.56$, $P < 0.001$) between the groups (Table 3).

Table 1

Frequency distribution of demographic variables of experimental and control groups.

Variable		Experimental group	Control group	Sig
		N (%)	N (%)	
Age (y)	60–63	16 (46)	15 (43)	$P = 0.6$
	64–67	13 (37)	16 (46)	
	68 years & older	6 (17)	4 (11)	
Marital status	Married	24 (69)	25 (72)	$P = 0.9$
	Single & widowed	11 (31)	10 (28)	
Educational level	Elementary	17 (49)	22 (63)	$P = 0.1$
	Middle & high school	15 (43)	7 (20)	
	Diploma & higher education	3 (7)	6 (17)	
Health insurance coverage	Yes	21 (60)	22 (63)	$P = 0.8$
	No	14 (40)	13 (37)	
Living with	Husband	23 (66)	25 (72)	$P = 0.9$
	Husband & children	1 (3)	1 (3)	
	Children	7 (20)	5 (14)	
Financial status	Stable	16 (46)	15 (43)	$P = 0.8$
	Unstable	19 (54)	20 (57)	
Family's income adequacy	Yes	13 (37)	9 (26)	$P = 0.1$
	No	8 (23)	4 (11)	
	To some extent	14 (40)	22 (63)	
Occupational status	Farmer & worker	9 (26)	14 (40)	$P = 0.3$
	Housewife & retiree	13 (37)	8 (23)	
	Employed after retirement	13 (37)	13 (37)	
BMI (kg/m ²)	25–22	9 (26)	8 (23)	$P = 0.2$
	26–29	15 (43)	21 (60)	
	30 and above	11 (31)	17 (47)	

Table 2

Comparison of the two groups' means and SD before and after intervention by ANCOVA analysis.

Variable	Phase	Experimental group	Control group	Group Sig.	F	η^2
SPA	Before	17.7 ± 3.2	16.5 ± 3.1	0.018	5.945	0.092
	After	14.6 ± 2.9	16.2 ± 2.5			
EDS	Before	29.2 ± 4.5	29.8 ± 4.7	<0.001	45.841	0.437
	After	21.2 ± 3.8	28.7 ± 5.5			
BMI	Before	28.1 ± 3.5	26.8 ± 2.2	0.002	10.690	0.189
	After	27.6 ± 3.4	27.3 ± 2.6			

η^2 = Partial Eta Squared.

3.5. BMI

The results represented that the BMI of experimental group before and after intervention was 28.1 ± 3.5 and 27.6 ± 3.4 , respectively. The

BMI of control group before and after intervention was 26.8 ± 2.2 and 27.3 ± 2.6 , respectively. As shown in Table 2, there was a significant difference in mean scores of BMI ($F = 10.69$, $p = 0.002$) between the groups.

4. Discussion

The aim of this study was to determine the effect of group walking program on SPA and the risk of EDs in aged women. One of the variables in this study was SPA. The mean score of SPA after intervention in experimental group decreased, which is consistent with the results of some study (Akyüz, 2017; Berger et al., 2019; Dogru, 2019; Gammage et al., 2016; McAuley et al., 2002). The result is inconsistent to another study (Portman et al., 2018). The reason for this difference may be related to the statistical population and inclusion criteria since the participants in their study were athletes and exercised approximately three times per week, but the subjects of the present study had no sport activities during the last six months. On the other hand, the women of the current study had SPA which was not an inclusion criterion in the Portman et al. study. The results of the ongoing study suggested significant differences in the mean score of SPA after intervention between two groups, and SPA in experimental group significantly decreased after the intervention. These findings are similar to those of other studies (Berger et al., 2019; Dogru, 2019).

Physical activity plays a decisive role in SPA. From the perspective of social psychology, if people feel that their bodies are remarkably disproportionate to those of their peers, they make a kind of social comparison, resulting in increased SPA and decreased motivation, but participation in physical activity with supporters such as spectators or those who exercise with the person reduces the SPA level (Babaei Bonab et al., 2019).

One of the other variables in this study was EDs. The EDs can cause disorders related to nutrition and threaten a person's health by changing food patterns and in taking poor nutrients (Hajihassani, 2019). In experimental group the mean score of EDs after intervention decreased, which is consistent with the results of some study (Hosseini Sarbazai et al., 2017; Mollazadeh Esfanjani et al., 2012; Pahlavan Yali & Hojjati Zidashti, 2016). The difference between two groups after intervention looks reasonably large and the results are statistically significant. Some results are consistent (Mollazadeh Esfanjani et al., 2012; Pahlavan Yali & Hojjati Zidashti, 2016) and this study is inconsistent with the findings of the present study (Hosseini Sarbazai et al., 2017).

Attitudes toward exercise play an important role in EDs (Mollazadeh Esfanjani et al., 2012). It seems that individuals with eating disorders are more likely to exercise for weight and shape reasons but less likely to exercise for enjoyment than health controls (Sala et al., 2017). Despite the high risk of somatic and psychological complications, it is estimated that about two-thirds of people with EDs seek treatment for reduced motivation, shame and fear of stigma (Pettersen et al., 2019). Exercise is commonly employed as a means of optimizing health and managing weight. However; not all individuals attain their desired long-term goals through exercise interventions (West et al., 2018). Unhealthy psychological and interpersonal factors seem involved in EDs and the treatment needs to long-term course (Rosenvinge & Pettersen, 2015) and a multidisciplinary approach, including medical, nutritional, social and

Table 3

Comparison of mean score of subscales of EDs post-test in experimental and control groups by MANCOVA.

Variable	Phase	Experimental group	Control group	Group Sig.	F	η^2
Dieting	Before	15.1 ± 2.8	15.4 ± 3.03	<0.001	29.896	0.344
	After	10.9 ± 2.4	14.7 ± 3.9			
Bulimia & food preoccupation	Before	6.5 ± 1.7	6.1 ± 1.6	0.007	7.688	0.119
	After	4.6 ± 1.8	5.9 ± 1.4			
Oral control	Before	7.5 ± 2.3	8.2 ± 1.4	<0.001	36.567	0.391
	After	5.7 ± 1.8	8.06 ± 1.4			

η^2 = Partial Eta Squared.

psychological components, and it needs to more studied (Podfigurna-Stopa et al., 2015).

In addition, the results of the ongoing study regarding the BMI of the experimental group before and after intervention are compatible with the results of a study conducted by Fantin et al. (2012) and Hosseini Sarbazai et al. (2017) and inconsistent with those of some studies (Berger et al., 2019; Dogru, 2019; Hornbuckle et al., 2016; Karstoft et al., 2013; Rodriguez-Hernandez & Wadsworth, 2019). The reason for this difference might be due to the differences in characteristics of samples and inclusion criteria as all aged women of the present study had SPA in addition to EDs. This anxiety increases the desire to do exercise to seem more attractive and well-formed (Babaei Bonab et al., 2019). In the present study there was a significant difference in mean scores of BMI between the groups that may be related to the small sample size. The results are inconsistent with those of some studies (Dogru, 2019; Hosseini Sarbazai et al., 2017; Rodriguez-Hernandez & Wadsworth, 2019).

Nevertheless, less activity is one of the main factors related to diseases; and exercise is one of the important and effective factors in improving the quality of life. Regular physical activity such as walking causes to reduce dependence on other family members and caregivers, improve mental and physical function as well as prevent premature death in middle-aged and aged people. However, those who do physical activity through walking will experience longer life spans and better quality of life (Shokravi et al., 2009). On the other hand, regarding the relationship between nature and human health that has been recognized since the earliest human civilizations in Iran, China, and Greece (Iwata et al., 2016), it may be argued that walking in nature causes a double effect of exercise on SPA and EDs in this research.

5. Strength of the study

We considered an easy intervention for the two problems on aged women on randomized clinical trial design with standard measures.

6. Study limitations

A notable limitation in this study was warm weather during intervention which caused some of the women to travel to the countryside.

7. Final conclusion

According to the results, the group walking program can be used as an attractive, affordable solution and easy approach has a positive effect on the aged women who have SPA and EDs.

CRedit authorship contribution statement

Sharare Khalili: Executor of the plan.
Fateme Shirinkam: Thesis guide, Proposal consuler.
Reza Ghadimi: Thesis guide, Statistical analyzer.
Hengameh Karimi: Thesis guide, Proposal designer and article manager.

Declaration of competing interest

There was no conflict in interest in the study.

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